Kevin Wei

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Education

University of Waterloo — Waterloo, ON
Major in Computer Engineering | GPA: 4.0 - Dean's Honour List — *Sep 2022 – Apr 2027*
Courses: Data Structures and Algorithms, Concurrency, OOP, Operating Systems, Databases, Compilers, Networks

Skills

Languages: Python, C, C++, Java, JavaScript, TypeScript, SQL
Frameworks/Libraries: Flask, FastAPI, Pandas, PyMongo, PyTorch, Spring Boot, React.js, Next.js, Node.js, Vue.js
Technologies: AWS, GCP, Azure, Kafka, PostgreSQL, MySQL, MongoDB, Terraform, Kubernetes, Docker, Selenium

Work Experience

AI Backend Developer — Jan 2025 - Apr 2025
Intact | Python, Flask, AWS, Kafka, Kubernetes, TensorFlow, Docker — *Toronto, ON*
- Architected a **Kafka-based microservice** orchestrating multi-layered extraction and **classification models** for routing claim documents, eliminating manual screening time by **20+ minutes** per document.
- Designed a robust document extraction system with **AWS** services including **Textract, SQS/SNS, DynamoDB, and S3**, processing over **5,000+** insurance claim documents daily with ensured data persistence.
- Developed an **AI system** with **agentic** capabilities that sources from business requirements, design documents, and meeting transcripts to create Jira tickets, reducing turnaround time by **70%**.
- Integrated NLP techniques such as **N-grams** and **fuzzy matching** to boost claim documents classification **accuracy by 9%** over legacy image-based models, cutting inference resource **cost by 80%**.

Research Assistant, Computer Vision — Sept 2025 - Dec 2025
University of Waterloo | PyTorch, OpenCV, NumPy — *Waterloo, ON*
- Leverage **deep learning** for object tracking to maintain persistent hockey player identities across video frames.
- Develop a camera-to-rink **homography** model with **CNN** to project 3D broadcast footage onto a normalized 2D rink map, enabling accurate player position mapping.

Software Developer, ML — May 2024 - Aug 2024
Chartd | Python, FastAPI, PyTorch, Pandas, AWS, Azure, Docker — *Ottawa, ON*
- Architected a **low-latency API pipeline** for real-time text processing, interpreting user instructions to trigger app functionalities, integrating sentiment analysis and RAG, and achieving **sub-2 second** response times.
- Developed a scalable infrastructure built on **AWS** for end-to-end model training and deployment, leveraging **SageMaker, S3, and ECR** to streamline data processing pipelines and reduce deployment time by **40%**.
- Delivered a custom data generation platform capable of producing datasets of up to **10 million** entries with adjustable input parameters, accelerating data preparation for **supervised fine-tuning** of LLMs by up to **60%**.

Full Stack Developer — Sep 2023 - Dec 2023
Chartd | Python, MongoDB, FastAPI, Pandas, Vue.js, TypeScript, Docker — *Ottawa, ON*
- Developed a Business Intelligence platform leveraging NLP models to analyze complex financial data.
- Implemented an ETL **data pipeline** for data transfers into MongoDB, achieving **sub-3 second** latency.
- Configured a router with **4 asynchronous endpoints** in FastAPI to batch process **10,000+ data nodes**.

Web Application Developer — Jan 2023 - Apr 2023
Chartd | Vue.js, Nuxt.js, TypeScript, HTML, Cypress — *Ottawa, ON*
- Developed Track Offsets, a financial management tool for businesses, abstracting data from JSON to TypeScript.
- Implemented CRUD interfaces used in **over 50 instances** within the app, using Cypress for **E2E Testing**.

Projects

ExpenseLink | *Python, React.js, Flask, JavaScript, Pandas, Google Cloud Platform (GCP)*
- Developed a mobile-friendly expense tracking app integrated with **Google Cloud Platform** and **Google Sheets**.
- Integrated persistent Google Login through Flask API with **OAuth 2.0 credentials**, using **stateless token-based authentication** with **HS256** encrypted tokens stored as JWTs in **Cookies**.

AI Vlog Platform | *Python, Flask, TypeScript, AWS, GCP*
- Built an **AI-driven** video editing platform, integrating **ShotStack API** for advanced editing capabilities.
- Utilised **AWS S3** for video storage and **Google Gemini** for automated video analysis and edit generation.